A Low-Latency Control Fabric for Distributed Quantum Error Correction using RDMA: A Timed Event Structure Model

Vasileios Klimis
Queen Mary University of London
UK
v.klimis@qmul.ac.uk

***Abstract*—Scalable, fault-tolerant quantum computing will likely rely on distributed architectures of multiple interconnected Quantum Processing Units (QPUs). The performance of such systems is critically limited by the latency of the classical communication required for feedback loops, especially within Quantum Error Correction (QEC) protocols. Traditional network stacks like TCP/IP introduce significant kernel-level overhead, creating latency that can easily exceed qubit coherence times and render QEC ineffective.**

In this paper, we propose a novel classical control plane architecture that leverages Remote Direct Memory Access (RDMA) to bypass the operating system kernel and enable ultra-low-latency communication between QPU control processors. To formally analyse this system, we introduce a Timed Event Structure model to precisely capture the causality and latency of the entire QEC feedback loop. Our analysis, based on this model, demonstrates that an RDMA-based fabric drastically reduces end-to-end latency compared to a TCP/IP-based approach, establishing it as a critical enabling technology for building robust, distributed, and fault-tolerant quantum computers.

***Index Terms*—Quantum Computing, Distributed Systems, Quantum Error Correction, RDMA, Low-Latency Networking, Formal Models.**

I. Introduction

The pursuit of a universal, fault-tolerant quantum computer faces a significant scalability challenge. While current quantum processors integrate tens to hundreds of qubits on a single chip, scaling to the millions of physical qubits required for meaningful applications presents immense engineering hurdles. A promising path forward lies in a distributed, modular architecture, where multiple smaller, high-quality QPU modules are interconnected to form a single, powerful computational resource. This approach mirrors the evolution of classical supercomputers from monolithic mainframes to clusters of commodity hardware [1].

However, this distributed paradigm introduces a new set of challenges, foremost among them being the coordination and control of quantum operations across modules. This is especially acute for Quantum Error Correction (QEC), the cornerstone of fault-tolerant quantum computation [2]. QEC protects fragile quantum states from decoherence by redundantly encoding logical qubits into many physical qubits and actively correcting errors as they occur. This process relies on a tight feedback loop: measure ancillary "syndrome" qubits to detect potential errors, classically communicate the measurement outcomes, decode the syndrome, and apply a corrective gate to the data qubits.

In a distributed system, the classical communication step becomes a network operation between the control processors of different QPU modules. The fundamental constraint of this process is that the total time for the feedback loop, T_{loop}, must be significantly shorter than the coherence time of the qubits, $T_{\text{coherence}}$. If this latency budget is exceeded, the quantum state will have decohered before a correction can be applied, defeating the purpose of QEC. Standard networking protocols like TCP/IP, with their inherent kernel-level processing, context switches, and buffer copies, introduce latencies that are often too high and unpredictable for this time-critical task [3], [4]. Studies have shown that traditional TCP sockets can exhibit latencies of 120 microseconds or more for small messages, while RDMA technologies can achieve sub-microsecond latencies [5].

This paper makes the following contributions to address this critical bottleneck:

1) We propose a novel system architecture for a distributed quantum control plane using an RDMA fabric.
2) We introduce a formal model based on Timed Event Structures to rigorously capture the causality and analyse the end-to-end latency of the distributed QEC feedback loop.
3) We provide a comparative analysis using this model, quantitatively demonstrating the substantial latency reduction of the RDMA-based approach over traditional networking, thereby establishing its feasibility for next-generation fault-tolerant systems.

The remainder of this paper is organised as follows. Section III provides background on QEC, classical interconnects, and our formal model. Section IV presents the proposed architecture and the event structure models. Section V analyses the results, and Section VII concludes the paper.

II. Related Work

The challenge of controlling large-scale quantum systems sits at the intersection of several research domains. Our work builds upon and contributes to three primary areas: quantum

979-8-3315-9092-5/25/$31.00 © 2025 IEEE
DOI 10.1109/PICom68402.2025.00043

control architectures, high-performance classical networking, and the formal modelling of concurrent systems.

1) Quantum Control Systems: Significant research has been dedicated to designing the classical hardware and software stack for controlling QPUs. Early and current systems typically employ a monolithic control architecture, where a powerful FPGA is tightly coupled to a single QPU, managing all real-time gate sequencing and measurement readout [6], [7]. These efforts have been instrumental in pushing qubit fidelity and coherence, focusing on minimising latency within a single, co-located system. While foundational, this body of work does not typically address the challenges of inter-module communication, where network latency, rather than on-chip signal propagation, becomes the dominant bottleneck. Recent work has begun exploring distributed architectures for quantum control [8], but the formal modelling of such systems remains limited. Our work explicitly extends this research into the distributed, multi-module paradigm.

2) High-Performance Interconnects: The benefits of kernel-bypass networking and RDMA are well-established in the classical high-performance computing (HPC) community [9], [10]. Technologies like InfiniBand and RoCE are standard in supercomputing clusters and data centres for applications like distributed databases and machine learning, where low latency and high throughput are critical. These studies provide extensive performance data for classical workloads, with RDMA technologies achieving sub-microsecond latencies compared to traditional TCP/IP networking. However, the application of these technologies to the unique, hard real-time constraints of a quantum feedback loop has been largely unexplored. The primary contribution of our paper is to bridge this gap by formally modelling and analysing the specific benefits of RDMA for the QEC use case.

3) Formal Methods for Quantum and Concurrent Systems: Formal methods have been increasingly applied to reason about quantum computation. Process algebras like the Communicating Quantum Processes (CQP) [11] and extensions of the quantum π-calculus [12] have been used to model and verify the correctness of quantum communication protocols. Other work has focused on the semantics of quantum programming languages. Our approach is complementary; rather than verifying the logical correctness of a protocol, we use Timed Event Structures, a classic formalism from concurrency theory, to model and analyse the *performance and causality* of the underlying physical and classical control infrastructure. This allows us to reason about timing, latency, and real-time feasibility, which are properties not typically captured by purely logical formalisms.

III. Background

To understand our contribution, we first provide essential background in both the quantum and networking domains.

A. The Quantum Error Correction (QEC) Feedback Loop

Quantum bits, or qubits, are extremely sensitive to environmental noise, which causes them to lose their quantum properties in a process called decoherence. QEC is a set of techniques designed to combat this. The core idea is to detect and correct errors without directly measuring (and thus destroying) the quantum state itself. This is achieved via a feedback cycle, illustrated conceptually in Fig. 1. An ancillary qubit, known as a syndrome qubit, is used to probe for errors on one or more data qubits. The syndrome qubit is then measured. The classical outcome of this measurement (e.g., '0' or '1') indicates the error syndrome. This classical information must be rapidly communicated to a control processor, which decodes it and issues a command to apply a specific corrective gate (e.g., a Pauli X or Z gate) to the data qubits. This entire cycle must repeat at a rate much faster than the decoherence rate.

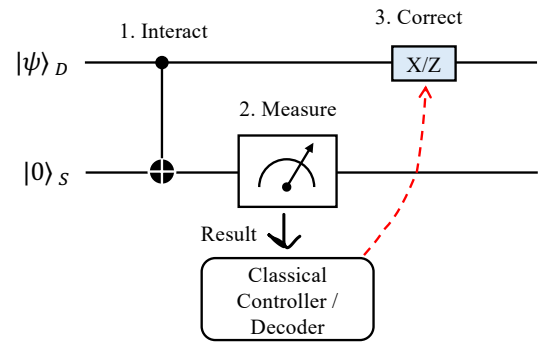

Fig. 1. A conceptual diagram of a single Quantum Error Correction (QEC) feedback loop. 1) A data qubit interacts with a syndrome qubit. 2) The syndrome qubit is measured, yielding a classical result. 3) This result is processed by a classical controller, which triggers a conditional corrective gate on the data qubit.

B. Classical Interconnects for High-Performance Systems

The classical feedback path shown in Fig. 1, from syndrome measurement to the application of a corrective gate, is the focus of our work.

1) TCP/IP: The standard networking protocol of the internet, TCP/IP, is designed for reliability over heterogeneous networks. However, its processing path involves the operating system kernel on both the sending and receiving ends. This incurs significant overhead from interrupts, system calls, and data copies between user space and kernel space, resulting in typical end-to-end latencies of tens to hundreds of microseconds (μs). This latency is often too high for the tight QEC timing budget.

2) Remote Direct Memory Access (RDMA): RDMA is a technology common in high-performance computing (HPC) and data centres [5], [13]. It allows a network adapter in one machine to directly read from or write to the memory of another machine, bypassing the remote CPU and OS kernel. This "kernel bypass" and the use of "one-sided" operations (`RDMA_Write`, `RDMA_Read`) dramatically reduces latency to just a few microseconds, while also minimising CPU overhead on the control processors [14], [15].

C. Formalism: Timed Event Structures

To reason formally about the performance of these systems, we use Timed Event Structures [16]. An event structure is a triple $(E, \leq, \#)$ where E is a set of events, $\leq$ is a partial order on E representing causality (i.e., $e_1 \leq e_2$ means e_1 must happen before e_2), and $\#$ is a symmetric, irreflexive relation representing conflict. For our analysis, we augment this with a timing model. We define a function time : $E \to \mathbb{R}^+$ that assigns a timestamp to the occurrence of each event, governed by the constraint that for any e_1, e_2 where e_1 is an immediate cause of e_2, we have $\text{time}(e_2) \geq \text{time}(e_1) + \delta$, where δ represents the minimum latency of the activity between e_1 and e_2.

IV. Modelling the Distributed QEC Loop

To enable a rigorous analysis of distributed QEC performance, and to precisely define and compare the end-to-end latency across different communication architectures, we model the system using *Timed Event Structures*, a formalism that extends the classical theory of Event Structures [16] by incorporating real-time constraints into partial-order models of concurrency [17].

A. System Architecture and Formal Components

We consider the distributed system architecture shown in Fig. 2, composed of multiple QPU modules. Each module is managed by a dedicated Local Control Processor (LCP). We define the LCP as the classical processing element responsible for the real-time control of its associated QPU, including the generation of precise gate control signals and the readout of measurement results. In practice, this LCP would typically be implemented using a Field-Programmable Gate Array (FPGA) or a custom System-on-a-Chip (SoC) to meet the stringent low-level timing requirements. Our model focuses on the interaction between a "Source" LCP, which measures a syndrome, and a "Corrector" LCP, which applies a corrective gate.

A key motivation for this distributed control architecture arises from the physical scale of advanced QEC codes. For powerful codes like the surface code, a single *logical qubit* may be encoded across hundreds of thousands of physical qubits, potentially spanning multiple QPU modules. To maintain the integrity of the code across these modules, *stabiliser measurements* must be performed on groups of qubits that include members from adjacent modules. This is achieved by using a local ancilla on a Source module to measure the joint state of data qubits on both its own module and a neighboring Corrector module, a process enabled by a short-range quantum interconnect. An error on a physical qubit in Module 2 will thus be detected by a measurement occurring on Module 1. The classical syndrome result is then decoded by LCP-1, which in turn sends a processed, *actionable command* – for instance, `(target_qubit_ID, corrective_gate_ID)` – via RDMA to LCP-2.

A Timed Event Structure is a tuple $\mathcal{E} = (E, \leq, \#, \delta)$, where:

- E is a set of events, representing actions in the system.

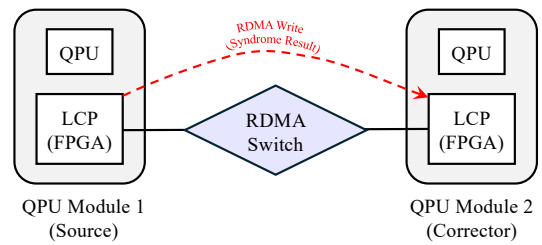

Fig. 2. Proposed RDMA-based Control Fabric Architecture.

- $\leq$ is a partial order on E representing causality. $e_1 \leq e_2$ means event e_1 is a necessary prerequisite for event e_2.
- $\#$ is a symmetric, irreflexive conflict relation. $e_1 \# e_2$ means events e_1 and e_2 are mutually exclusive choices in a single run of the system.
- $\delta : E \times E \to \mathbb{R}^+$ is a latency function, where $\delta(e_i, e_j)$ defines the minimum time delay between an event e_i and an immediately succeeding event e_j.

Two events e_1, e_2 are defined as *concurrent* (e_1 co e_2) if they are not ordered and not in conflict, i.e., $\neg(e_1 \leq e_2)$, $\neg(e_2 \leq e_1)$, and $\neg(e_1 \# e_2)$. A *causal chain* is a sequence of events $\langle e_1, e_2, \ldots, e_n \rangle$ such that $e_1 \leq e_2 \leq \cdots \leq e_n$.

B. Event Structure for a TCP/IP-based Loop (Baseline)

The event structure for the TCP/IP-based feedback loop, $\mathcal{E}_{\text{TCP}}$, captures the full software and hardware stack traversal shown in Fig. 3.

The set of events, E_{TCP}, is defined as:

- e_{meas}: Syndrome measurement completes at the Source LCP.
- $e_{\text{send_app}}$: The source application invokes the 'send()' system call.
- $e_{\text{kern_send}}$: The source OS kernel completes TCP/IP processing.
- $e_{\text{net_tx}}$: The physical transmission across the network fabric completes.
- $e_{\text{kern_recv}}$: The destination OS kernel completes packet processing.
- $e_{\text{app_recv}}$: The data is delivered to the destination application.
- e_{decode}: The destination LCP decodes the error syndrome.
- e_{correct}: The corrective gate is applied.

In this model, the feedback loop constitutes a single, long causal chain. The causality relation $\leq_{\text{TCP}}$ is a total order on this subset of events:

$$e_{\text{meas}} \leq e_{\text{send_app}} \leq \cdots \leq e_{\text{decode}} \leq e_{\text{correct}}$$

For this linear process, there is no concurrency and we assume no conflict. The end-to-end latency is formally defined as the sum of latencies along this maximal causal chain:

$$T_{\text{loop}}^{\text{TCP}} = \sum_{i=1}^{n-1} \delta(e_i, e_{i+1})$$

This formal definition directly corresponds to the summation presented in our analysis in Section V.

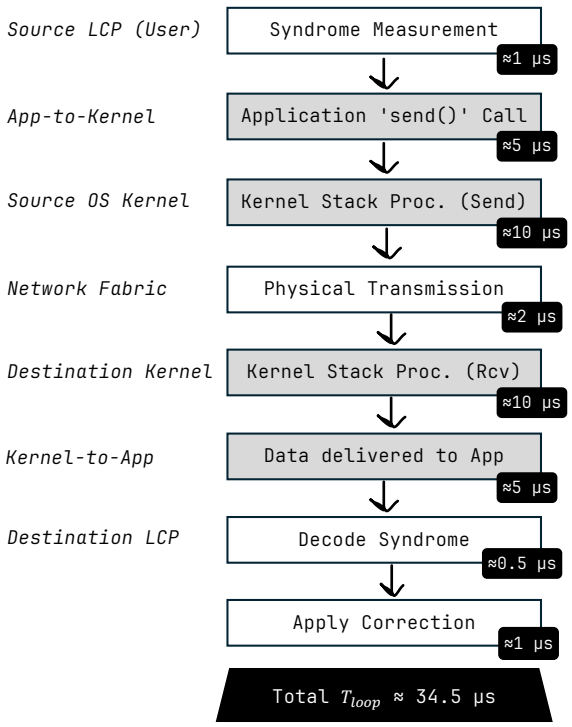

Fig. 3. Timed Event Structure for the TCP/IP-based Loop.

C. Event Structure for an RDMA-based Loop (Proposed)

The RDMA-based architecture yields a fundamentally simpler event structure, $\mathcal{E}_{\text{RDMA}}$, as depicted in Fig. 4. The kernel-level events are eliminated entirely.

The set of events, E_{RDMA}, is significantly smaller:

- e_{meas}: Syndrome measurement completes.
- $e_{\text{rdma_w}}$: The source LCP initiates the `RDMA_Write` operation.
- $e_{\text{rdma_c}}$: The data arrives in the destination LCP's memory.
- e_{decode}: The destination LCP decodes the syndrome.
- e_{correct}: The corrective gate is applied.

The causality relation $\leq_{\text{RDMA}}$ forms a much shorter causal chain:

$$e_{\text{meas}} \leq e_{\text{rdma_w}} \leq e_{\text{rdma_c}} \leq e_{\text{decode}} \leq e_{\text{correct}}$$

The end-to-end latency is again the sum of latencies along this chain:

$$T_{\text{loop}}^{\text{RDMA}} = \sum_{j=1}^{m-1} \delta(e'_j, e'_{j+1})$$

The formal comparison between $\mathcal{E}_{\text{TCP}}$ and $\mathcal{E}_{\text{RDMA}}$ makes the architectural advantage explicit: the set E_{RDMA} has a smaller cardinality than E_{TCP}, and its maximal causal chain is shorter. This structural difference in the models directly proves why $T_{\text{loop}}^{\text{RDMA}} \ll T_{\text{loop}}^{\text{TCP}}$.

While our analysis focuses on a single feedback loop, this event structure formalism is highly expressive. For instance, if the classical controller could trigger one of several corrective gates based on the syndrome, these alternative e_{correct} events would be modelled using the conflict relation (#). Similarly, if multiple QEC loops were running in parallel on different modules, their respective event chains would be concurrent (co) in the global event structure, providing a formal basis for analysing system-wide parallelism and resource contention.

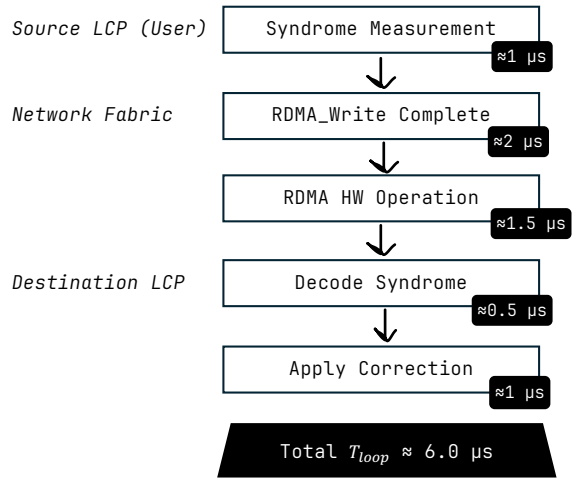

Fig. 4. Timed Event Structure for the RDMA-based Loop.

V. Analysis and Discussion

Having defined the formal models, we now use them to perform a quantitative analysis. The primary metric of interest is the end-to-end latency, T_{loop}, calculated from the initial event e_{meas} to the final event e_{correct}.

One might ask why a formal model like event structures is necessary when the feedback loop appears to be a simple, linear sequence of operations whose delays could be summed using basic arithmetic. For this isolated, single-loop case, that is true. However, the true value of the event structure formalism lies in its ability to rigorously model the *concurrency, conflict, and complex causal dependencies* inherent in a realistic, large-scale distributed quantum computer. We introduce it here to build a foundation that is extensible to these more complex scenarios.

For example, consider a system executing multiple QEC loops simultaneously on different QPU modules. The events of one loop would be formally *concurrent* (e_A co e_B) with the events of another. This allows us to reason about system-wide properties like parallelism, throughput, and potential resource contention on the shared network fabric. A simple linear calculation cannot capture these concurrent interactions.

Furthermore, the conflict relation (#) in event structures is essential for modelling non-determinism and choice. If a syndrome decoding could result in several mutually exclusive corrective actions (e.g., apply gate X or gate Y but not both), these would be modelled as conflicting events. This provides a formal basis for analysing the timing properties of all possible execution paths in a non-deterministic protocol.

Therefore, while our initial application of the model is to a linear chain, we have deliberately chosen a formalism that is not limited to this simple case. It provides the necessary expressive power to serve as a robust framework for analysing

TABLE I
COMPARATIVE LATENCY BREAKDOWN FOR A SINGLE QEC FEEDBACK LOOP

Stage	TCP/IP Latency (μs)	RDMA Latency (μs)
Syndrome Measurement	1.0	1.0
Control Logic / App-to-Kernel	5.0	-
Kernel Stack Processing (Send)	10.0	-
Network Fabric (HW)	2.0	2.0
Kernel Stack Processing (Rcv)	10.0	-
Kernel-to-App / Interrupt	5.0	-
RDMA HW Operation	-	1.5
Syndrome Decode	0.5	0.5
Corrective Gate Application	1.0	1.0
Total T_{loop}	**34.5 μs**	**6.0 μs**

the complex, concurrent, and non-deterministic behaviour of future fault-tolerant quantum systems.

With this justification, we now proceed with the quantitative analysis of the two architectures. The total latency is the sum of minimum time delays, $\delta(e_i, e_j)$, along a causal chain. The component latencies, derived from established architectural principles, are summarised in Table I.

The analysis clearly shows that the RDMA-based approach reduces the classical communication latency by nearly an order of magnitude. This has profound implications for fault tolerance. While the average loop latency (T_{loop}) provides a stark comparison, it only tells part of the story. Fault-tolerant quantum computation is a real-time process where timing predictability is as crucial as speed. The latency of a TCP/IP-based system exhibits high variance, or **jitter**, due to unpredictable software phenomena such as OS scheduler pre-emption, interrupt coalescing, and network buffer contention. A single delayed packet could cause a QEC cycle to miss its deadline, invalidating the correction.

The RDMA architecture, by offloading the communication path to dedicated hardware, provides not only lower average latency but also significantly lower jitter. The hardware-based path is deterministic and not subject to the scheduling vagaries of a general-purpose OS kernel. This high degree of predictability is essential for building a synchronous and reliable control system capable of executing millions of QEC cycles with consistent timing.

Consider a QPU with a coherence time $T_{\text{coherence}}$ of 200 μs. The number of QEC cycles possible within this time for each model is:

- **TCP/IP Cycles:** $200\,\mu\text{s}/34.5\,\mu\text{s} \approx 5$ cycles
- **RDMA Cycles:** $200\,\mu\text{s}/6.0\,\mu\text{s} \approx 33$ cycles

The ability to perform over six times as many error correction cycles within the same coherence window dramatically improves the probability of catching and correcting errors before they corrupt the logical state. This translates directly to a lower logical error rate, making the RDMA fabric a key enabler for robust, distributed fault-tolerant quantum computation. Furthermore, as systems scale to more modules, the predictable, low-latency performance of RDMA becomes even more critical for managing complex, system-wide error syndromes.

VI. CHALLENGES AND OPEN RESEARCH DIRECTIONS

While our work establishes the critical role of a low-latency classical fabric, it also opens several avenues for future research at the intersection of quantum and distributed systems. We identify the following key challenges:

1) Scalability of Classical Decoding: This paper addresses the communication latency, but for larger, more powerful QEC codes like the surface code, the classical computation required to *decode* a complex syndrome can itself become a significant bottleneck. A major open challenge is the co-design of the low-latency fabric with high-throughput, parallel classical decoders to ensure that computation does not become the new limiting factor after communication has been solved.

2) Global Synchronisation and Timestamping: Coordinating operations across dozens or hundreds of distributed QPU modules requires a globally consistent and highly precise sense of time. Integrating precision timing protocols, such as PTP (Precision Time Protocol), over an RDMA fabric to provide sub-microsecond synchronisation to all classical controllers is a non-trivial distributed systems problem. The interplay between the network's timing guarantees and the quantum algorithm's coherence requirements is a rich area for investigation.

3) Resource Management for Pinned Memory: A prerequisite for RDMA is that the memory regions used for communication must be "pinned" or "registered" with the network hardware, preventing the OS from moving or swapping them out. In a dynamic quantum computing environment where workloads may change, managing these pinned memory resources efficiently becomes a complex software engineering challenge. New resource management frameworks will be needed to allocate, register, and reclaim these memory regions without impacting real-time performance.

VII. CONCLUSION

The scalability of quantum computers from the current NISQ era to future fault-tolerant systems hinges on the development of modular, distributed architectures. In this paper, we have addressed a critical and often underestimated bottleneck in such systems: the high latency and unpredictability of the classical communication fabric required for real-time Quantum Error Correction. We identified that traditional network protocols like TCP/IP, with their inherent software overhead, are fundamentally ill-suited for the microsecond-scale feedback loops demanded by QEC.

To solve this, we proposed a novel control plane architecture based on a high-performance, kernel-bypass RDMA fabric. By applying Timed Event Structures as a formal modelling tool, we rigorously analysed and compared our proposed architecture against a conventional TCP/IP-based system. Our analysis yielded three key findings. First, the RDMA-based approach reduces the end-to-end feedback loop latency by nearly an order of magnitude. Second, by offloading communication to dedicated hardware, it provides the low-jitter, deterministic timing predictability that is essential for synchronous real-time control. Third, as a consequence, our proposed architecture

enables a dramatically higher number of corrective cycles within a given qubit coherence time, directly enhancing the efficacy of QEC.

This work establishes that the classical interconnect is not a mere implementation detail but a foundational component of a fault-tolerant quantum computer. Just as high-performance networking enabled the rise of classical distributed computing, low-latency, kernel-bypass fabrics like RDMA will be a key enabling technology for the future of distributed quantum computing. The principles and models presented here provide a framework for designing and reasoning about the performant and reliable control systems that will be necessary to unlock the full potential of large-scale quantum machines.

References

[1] J. Dongarra, P. Beckman *et al.*, "The international exascale software project roadmap," *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, vol. 369, no. 1949, pp. 3412–3435, 2011.

[2] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th ed. Cambridge University Press, 2010.

[3] P. Balaji, H. V. Shah, and D. K. Panda, "Sockets vs. rdma interface over 10-gigabit networks: An in-depth analysis of the memory traffic bottleneck," 2004. [Online]. Available: https://api.semanticscholar.org/CorpusID:2598751

[4] Y. Wan, D. Feng, F. Wang, L. Ming, and Y. Xie, "An in-depth analysis of tcp and rdma performance on modern server platform," in *2012 IEEE Seventh International Conference on Networking, Architecture, and Storage*, 2012, pp. 164–171.

[5] C. Mitchell, Y. Geng, and J. Li, "Using one-sided rdma reads to build a fast, cpu-efficient key-value store," in *2013 USENIX Annual Technical Conference*, 2013, pp. 103–114.

[6] Y. Yang, Z. Shen, X. Zhu, Z. Wang, G. Zhang, J. Zhou, X. Jiang, C. Deng, and S. Liu, "Fpga-based electronic system for the control and readout of superconducting quantum processors," *Review of Scientific Instruments*, vol. 93, no. 7, Jul. 2022. [Online]. Available: http://dx.doi.org/10.1063/5.0085467

[7] Y. Xu, G. Huang, J. Balewski, R. Naik, A. Morvan, B. Mitchell, K. Nowrouzi, D. I. Santiago, and I. Siddiqi, "Qubic: An open-source fpga-based control and measurement system for superconducting quantum information processors," *IEEE Transactions on Quantum Engineering*, vol. 2, p. 1–11, 2021. [Online]. Available: http://dx.doi.org/10.1109/TQE.2021.3116540

[8] N. Fruitwala, G. Huang, Y. Xu, A. Rajagopala, A. Hashim, R. K. Naik, K. Nowrouzi, D. I. Santiago, and I. Siddiqi, "Distributed architecture for fpga-based superconducting qubit control," 2024. [Online]. Available: https://arxiv.org/abs/2404.15260

[9] A. Kalia, M. Kaminsky, and D. G. Andersen, "Design guidelines for high performance rdma systems," *Proceedings of the 2016 USENIX Annual Technical Conference*, pp. 437–450, 2016.

[10] NVIDIA Corporation, "Infiniband and roce performance analysis," NVIDIA, Tech. Rep., 2023, technical Documentation.

[11] S. J. Gay and R. Nagarajan, "Communicating quantum processes," *SIGPLAN Not.*, vol. 40, no. 1, p. 145–157, 2005. [Online]. Available: https://doi.org/10.1145/1047659.1040318

[12] S. J. Gay and I. V. Puthoor, "Application of quantum process calculus to higher dimensional quantum protocols," *Electronic Proceedings in Theoretical Computer Science*, vol. 158, p. 15–28, Jul. 2014. [Online]. Available: http://dx.doi.org/10.4204/EPTCS.158.2

[13] A. Dragojević, D. McKenzie, D. Narayanan, and M. Castro, "Rdma reads: To use or not to use?" *IEEE Data Engineering Bulletin*, vol. 37, no. 3, pp. 90–96, 2014.

[14] T. Ziegler, V. Leis, and C. Binnig, "Rdma communciation patterns," *Datenbank-Spektrum*, pp. 199–210, 2020.

[15] X. Kong, J. Chen, W. Bai, Y. Xu, M. Elhaddad, S. Raindel, J. Padhye, A. R. Lebeck, and D. Zhuo, "Understanding RDMA microarchitecture resources for performance isolation," in *20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23)*. Boston, MA: USENIX Association, Apr. 2023, pp. 31–48. [Online]. Available: https://www.usenix.org/conference/nsdi23/presentation/kong

[16] G. Winskel, "An introduction to event structures," *Lecture Notes in Computer Science*, vol. 354, pp. 364–397, 1989.

[17] I. B. Virbitskaite and R. S. Dubtsov, "Semantic domains of timed event structures," *Program. Comput. Softw.*, p. 125–137, May 2008. [Online]. Available: https://doi.org/10.1134/S0361768808030018